

## Chapter 5, Section 3

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1. Let  $X$  be a nonsingular projective variety of any dimension, let  $Y$  be a nonsingular subvariety, and let  $\pi : \tilde{X} \rightarrow X$  be obtained by blowing up  $Y$ . Show that  $p_a(\tilde{X}) = p_a(X)$ .

*Proof.* The proof of (3.4) follows through for the more general case by (II, 8.24).

**Proposition 0.1.** *In the above setting, we have  $\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ , and  $R^i \pi_* \mathcal{O}_{\tilde{X}} = 0$  for  $i > 0$ . Therefore,  $H^i(X, \mathcal{O}_X) \cong H^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$  for all  $i \geq 0$ .*

□

2. Let  $C$  and  $D$  be curves on a surface  $X$ , meeting at a point  $P$ . Let  $\pi : \tilde{X} \rightarrow X$  be the monoidal transformation with center  $P$ . Show that  $\tilde{C} \cdot \tilde{D} = C \cdot D - \mu_P(C) \cdot \mu_P(D)$ . Conclude that  $C \cdot D = \sum \mu_P(C) \cdot \mu_P(D)$ , where the sum is taken over all intersection points of  $C$  and  $D$ , including infinitely near intersection points.

*Proof.* Let  $E$  be the exceptional locus of  $\pi : \tilde{X} \rightarrow X$ . By (3.6), we have

$$\pi^* C = \tilde{C} + \mu_P(C)E.$$

Hence, we have

$$\tilde{C} \cdot \tilde{D} = (\pi^* C - \mu_P(C)E) \cdot (\pi^* D - \mu_P(D)E) = C \cdot D - \mu_P(C) \cdot \mu_P(D).$$

Assuming  $C$  and  $D$  are transversal, meeting at a finite number of points, the strict transforms of  $C$  and  $D$  under the blowing up of all intersection points will not meet, i.e.,  $\tilde{C} \cdot \tilde{D} = 0$ . Each time an intersection point is blown up,  $C \cdot D$  goes down by the product of the intersection multiplicities of each curve. □

3. Let  $\pi : \tilde{X} \rightarrow X$  be a monoidal transformation, and let  $D$  be a very ample divisor on  $X$ . Show that  $2\pi^* D - E$  is ample on  $\tilde{X}$ .

*Proof.*

□

5. Let  $a_1, \dots, a_r$ ,  $r \geq 5$ , be distinct elements of  $k$ , and let  $C$  be the curve in  $\mathbb{P}^2$  given by the (affine) equation  $y^2 = \prod_{i=1}^r (x - a_i)$ . Show that the point  $P$  at infinity on the  $y$ -axis is a singular point. Compute  $\delta_P$  and  $g(\tilde{Y})$ , where  $\tilde{Y}$  is the normalization of  $Y$ . Show in this way that one obtains hyperelliptic curves of genus  $g \geq 2$ .

7. For each of the following singularities at  $(0, 0)$  in the plane, give an embedded resolution, compute  $\delta_P$ , and decide which ones are equivalent.

- (a)  $x^3 + y^5 = 0$
- (b)  $x^3 + x^4 + y^5 = 0$
- (c)  $x^3 + y^4 + y^5 = 0$
- (d)  $x^3 + y^5 + y^6 = 0$
- (e)  $x^3 + xy^3 + y^5 = 0$